GANITA PRAKASH

CLASS 7 MATHEMATICS | PART I

ARITHMETIC EXPRESSIONS

Solved textbook exercises + illustrated notes

$$4 \times (8 + 3)$$
$$= 4 \times 8 + 4 \times 3$$
$$= 44$$

Questions first → worked explanations → visual notes → revision

Aligned with the supplied NCERT *Ganita Prakash*, Grade 7, Part I.
Original explanations and original TikZ illustrations; editable in Overleaf.

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Textbook exercises: worked answers first

Follow the reasoning

These answers come before the lesson notes. For additional textbook questions the full provided solution transcript follows, with original textbook page references. A good answer shows the operation used, *not just* the final number.

Textbook question + answer

Textbook p. 24–25. Express 24 in four ways. Fill the equalities $13 + 4 = \square + 6$, $22 + \square = 6 \times 5$, $8 \times \square = 64 \div 2$, $34 - \square = 25$. **Answer:** $24 = 12 + 12 = 30 - 6 = 4 \times 6 = 48 \div 2$. Blank values are respectively $11, 8, 4, 9$.

Textbook question + answer

Textbook p. 25. Arrange in ascending order: $67 - 19$, $67 - 20$, $35 + 25$, 5×11, $120 \div 3$. **Answer:** Values: $48, 47, 60, 55, 40$. Therefore $120 \div 3 < 67 - 20 < 67 - 19 < 5 \times 11 < 35 + 25$.

Textbook question + answer

Textbook p. 25–26. Compare expressions without heavy calculation. **Answer:** $1023 + 125 = 1148 < 1150 = 1022 + 128$. $113 - 25 = 88 = 112 - 24$. Idea: compare the *changes* to the terms instead of recomputing everything.

Textbook question + answer

Textbook p. 28–31. Why is subtraction treated as addition of the opposite? Can we rearrange terms? **Answer:** $18 - 10 = 18 + (-10) = 8$. For a sum written as signed terms, rearrangement does not change its value: $9 + (-4) + 6 = 9 + 6 + (-4) = 11$. But you cannot change a subtraction expression by exchanging numbers *without keeping each term's sign.*

Textbook question + answer

Textbook p. 37–38. Compare two different ways of using brackets. **Answer:** $73 - 14 + 1 = 60 = 73 - (14 - 1) = 73 + (-14 + 1)$. But $73 - 14 - 1 = 58 = 73 - (14 + 1) = 73 + (-14 - 1)$.

Textbook question + answer

Textbook p. 39–41. Multiply with the distributive property. **Answer:** $5 \times (4+3) = 5 \times 4 + 5 \times 3 = 35$, whereas $5 \times 4 + 3 = 23$. Also $(100 - 5) \times 8 = 760$; $(100 + 4) \times 15 = 1560$; $(50 - 1) \times 50 = 2450$.

Textbook question + answer

Textbook p. 41–42. Fill a distributive identity and match equivalent forms. **Answer:** $3(5 + 8) = 3 \times 5 + 3 \times 8$. $(15 - 6) \times 7 = 15 \times 7 - 6 \times 7$. $(17 - 9) \times 7 = 17 \times 7 - 9 \times 7$. Multiplication must be distributed to **both** terms inside the brackets.

Textbook question + answer

Textbook p. 42–43. Rahim supplies 9 kg mangoes and Shyam 11 kg each day for 7 days. Binu's annual savings. **Answer:** Mangoes: $7(9+11) = 140$ kg. Savings: $12[20{,}000 - (5000 + 5000 + 2000)] = 12 \times 8000 = \text{Rs. } 96{,}000$.

Textbook question + answer

Textbook p. 43. A snail climbs 3 cm in daytime and slips 2 cm at night. Reach a 10 cm post? **Answer:** After seven whole days it is at 7 cm. On **day 8**, it climbs from 7 cm to 10 cm and reaches the top. It does *not* slip again after reaching the treat.

Textbook question + answer

Textbook p. 43. Melvin reads 1 story per reading day, five days a week for eight weeks. **Answer:** $5 \times 8 = 40$ stories. Both $(7 - 2) \times 8$ and $7 \times 8 - 2 \times 8$ represent it. His stories each span two pages, but the unit asked is *stories*, not pages.

Textbook question + answer

Textbook p. 43–44. Simplify alternating sums and identify equal expressions. **Answer:** $1 - 2 + 3 - 4 + 5 - 6 + 7 - 8 + 9 - 10 = -5$. $1 - 1 + 1 - 1 + \cdots + 1 - 1 = 0$ (five pairs). $83 - 37 - 12$ equals both $84 - 38 - 12$ and $-37 + 83 - 12$.

Common Mistake

Parentheses are not decoration: $5(4 + 3) = 35$ but $5 \times 4 + 3 = 23$. In $-4 + 6$, the term -4 carries its minus sign when terms are regrouped.

Further textbook exercise solutions (editable text)

How to use this companion answer section

The supplied chapter PDF contains *additional solution pages after the textbook chapter*. The following is an **editable transcription** organised in that same order, so that page-based questions are not lost. Equations originally set as stacked fractions, tables, and picture puzzles may lose some spatial formatting in automatic text extraction. **For those, consult the original supplied PDF figure**; the worked explanations above and the visual notes later are the clearest first reference. Answers in the supplied companion have not all been independently verified.

More textbook solutions — companion p. 1

Page No. 24 Choose your favourite number and write as many expressions as you can having that value.

Answer: Let us choose the number 24. The arithmetic expressions for the number can be as follows: 12 + 12 = 24; 4 × 6 = 24; 48 ÷ 2 = 24, 34 – 10 = 24, 20 + 4 = 24 etc. Now you choose your number and form arithmetic expressions for that number.

Page No. 25

Figure it Out

1. Fill in the blanks to make the expressions equal on both sides of the = sign: (a) 13 + 4 = _____ + 6 (b) 22 + _____ = 6 × 5 (c) 8 × _____ = 64 ÷ 2 (d) 34 – _____ = 25

Answer: 1. (a) 11 (b) 8 (c) 4 (d) 9

2. Arrange the following expressions in ascending (increasing) order of their values. (a) 67 – 19 (b) 67 – 20 (c) 35 + 25 (d) 5 × 11 (e) 120 ÷ 3

Answer: 2. 120 ÷ 3 < 67 – 20 < 67 – 19 < 5 × 11 < 35 + 25

Page No. 26 Use '>' or '<' '=' in each of the following expressions to compare them. Can you do it without complicated calculations? Explain your thinking in each case.

Answer: (a) > (b) = (c) < (d) < (e) <

More textbook solutions — companion p. 2

Page No. 28 – 29 Check if replacing subtraction by addition in this way does not change the value of the expression, by taking different examples.

Answer: Let us take the numbers. 18 and 10 18 – 10 = 8 and 18 + (– 10) = 8 Can you explain why subtracting a number is the same as adding its inverse, using the Token Model of integers that we saw in the Class 6 textbook of mathematics?

Answer: Refer chapter 10 of class VI Mathematics Textbook. Does changing the order in which the terms are added give different values?

Answer: No, For example, 14 + 10 +(– 5) = 19 or (– 5) + 10 + 14 = 19

Page No. 29 Will this also hold when there are terms having negative numbers as well? Take some more expressions and check.

Answer: Yes, For example: (– 4) + (– 2) = – 6 or (– 2) + (– 4) = – 6 (– 6) + (– 8) = – 14 or (– 8) + (– 6) = – 14 Write some more examples.

Page No. 30 Can you explain why this is happening using the Token Model of integers that we saw in the Class 6 textbook of mathematics?

More textbook solutions — companion p. 3

Answer: Yes For example, Other way Can you explain why this is happening using the Token Model of integers that we saw in the Class 6 textbook of mathematics?

Answer: Yes. Consider the expression: $-6 + (-7) + (-13)$

Page No. 31 Does adding the terms of an expression in any order give the same value? Take some more expressions and check. Consider expressions with more than 3 terms also.

Answer: Yes, adding the terms of an expression in any order gives the same value Consider an expression with 3 terms $9 + 4 + 6$ Order 1: $9 + 4 + 6 = (9 + 4) + 6 = 13 + 6 = 19$ Order 2: $9 + 4 + 6 = 9 + (4 + 6) = 9 + 10 = 19$ An expression with 4 terms (including a negative number) Consider the expression: $15 + (-8) + 5 + 12$ Order 1: $(15 + (-8)) + 5 + 12 = (7 + 5) + 12 = 12 + 12 = 24$ Order 2: $15 + 5 + 12 + (-8) = (15 + 5 + 12) + (-8)$

More textbook solutions — companion p. 4

Answer: No, she can add fourth number, 9055 to the sum she got (11749) to get the correct sum of the list of given numbers. That is $11749 + 9055 = 20804$ (Using Associative Property)

Page No. 32 If the total number of friends goes up to 7 and the tip remains the same, how much will they have to pay? Write an expression for this situation and identify its terms.

Answer: The expression for the total cost $= 7 \times 23 + 5 = 161 + 5 =$ Rs. 166. The terms in the expression $7 \times 23 + 5$ are 7×23, 5.

Page No. 33 For each of the cases below, write the expression and identify its terms: (a) If the teacher had called out '4', Ruby would write _____ (b) If the teacher had called out '7', Ruby would write _____ Write an expression like the above for your class size.

Answer: (a) $8 \times 4 + 1$ Terms: 8×4, 1 (b) $4 \times 7 + 5$ Terms: 4×7, 5

More textbook solutions — companion p. 5

Answer: Way 1: $432 = 4 \times 100 + 1 \times 20 + 1 \times 10 + 2 \times 1$ Terms: 4×100, 1×20, 1×10, and 2×1 Way 2: $432 = 8 \times 50 + 1 \times 10 + 4 \times 5 + 2 \times 1$ Terms: 8×50, 1×10, 4×5, and 2×1 Can you think of some more ways of giving Rs. 432 to someone?

Answer: $40 \times 10 + 3 \times 10 + 2 \times 1$ Terms: 40×10, 3×10 and 2×1 Try for some more expressions.

Page No. 34 – 35

Figure it Out

1. Find the values of the following expressions by writing the terms in each case. (a) $28 - 7 + 8$ (b) $39 - 2 \times 6 + 11$ (c) $40 - 10 + 10 + 10$ (d) $48 - 10 \times 2 + 16 \div 2$ (e) $6 \times 3 - 4 \times 8 \times 5$

Answer: (a) $28 - 7 + 8$ Terms: 28, -7, and 8 Value: 29 (b) $39 - 2 \times 6 + 11$ Terms: 39, -2×6 and 11 Value: 38 (c) $40 - 10 + 10 + 10 = 40 + (-10) + 10 + 10$ Terms: 40, -10, 10 and 10 Value: 50 (d) $48 - 10 \times 2 + 16 \div 2 = 48 + (-10 \times 2) + (16 \div 2)$ Terms: 48, -10×2, $16 \div 2$ Value: 36 (e) $6 \times 3 - 4 \times 8 \times 5 = (6 \times 3) + (-4 \times 8 \times 5)$ Terms: 6×3, $4 \times 8 \times 5$ Value: -142

More textbook solutions — companion p. 6

2. Write a story/situation for each of the following expressions and find their values. () $89 + 21 - 10$ (b) $5 \times 12 - 6$ (c) $4 \times 9 + 2 \times 6$

Answer: (a) $89 + 21 - 10$; One of the stories could be; A library had 89 books on its shelves. The librarian bought 21 new books and added them to the collection. Later, 10 books were borrowed by students. How many books remain on the shelves? Value of the expression: 100 (b) $5 \times 12 - 6$ Make a story of your own for the given expression. Value of the expression: 54 (c) $4 \times 9 + 2 \times 6$ Make a story of your own for the given expression. Value of the expression: 48

3. For each of the following situations, write the expression describing the situation,

identify its terms, and find the value of the expression. () Queen Alia gave 100 gold coins to Princess Elsa and 100 gold coins to Princess Anna last year. Princess Elsa used the coins to start a business and doubled her coins. Princess Anna bought jewellery and has only half of the coins left. Write an expression describing how many gold coins Princess Elsa and Princess Anna together have.

More textbook solutions — companion p. 7

Answer: (a) Expression describing the situation = 2 × 100 + 100 2 Terms: 2 × 100, 100 2 Value: 250 gold coins. (b) (i) Expression = 4 × 40 + 3 × 20 Terms: 4 × 40, 3 × 20 Value: Rs. 220 (ii) Expression = 2 × (3 × 40). Terms: 2 × (3 × 40) Value: Rs. 240 (c) Expression = 7 × 5 + 6 × 2 + 2 × 3 Terms: 7 × 5, 6 × 2, 2 × 3 Total height: 53 cm

Page No. 37 Some expressions are given in the following three columns. In each column, one or more terms are changed from the first expression. Go through the example (in the first column) and fill in the blanks, doing as little computation as possible.

More textbook solutions — companion p. 8

More textbook solutions — companion p. 9

Page No. 37 – 38

Figure it Out

1. Fill in the blanks with numbers, and boxes with operation signs such that the expressions on both sides are equal. (a) 24 + (6 – 4) = 24 + 6 _____ (b) 38 + (_____ _____) = 38 + 9 – 4 (c) 24 – (6 + 4) = 24 6 – 4 (d) 24 – 6 – 4 = 24 – 6 _____ (e) 27 – (8 + 3) = 27 _____ 8 _____ 3 (f) 27 – (_____ _____) = 27 – 8 + 3

Answer: (a) 24 + (6 – 4) = 24 + 6 4 (b) 38 + (9 4) = 38 + 9 – 4 (c) 24 – (6 + 4) = 24 6 – 4 (d) 24 – 6 – 4 = 24 – 6 4 (e) 27 – (8 + 3) = 27 – 8 – 3 (f) 27 – (8 3) = 27 – 8 + 3

2. Remove the brackets and write the expression having the same value. (a) 14 + (12 + 10) (b) 14 – (12 + 10) (c) 14 + (12 – 10) (d) 14 – (12 – 10) (e) – 14 + 12 – 10 (f) 14 – (– 12 – 10)

Answer: (a) and (f)

3. Find the values of the following expressions. For each pair, first try to guess whether they have the same value. When are the two expressions equal? (a) (6 + 10) – 2 and 6 + (10 – 2) (b) 16 – (8 – 3) and (16 – 8) – 3 (c) 27 – (18 + 4) and 27 + (– 18 – 4)

More textbook solutions — companion p. 10

Answer: (a) (6 + 10) – 2 = 6 + 10 – 2 = 14 6 + (10 – 2) = 6 + 10 – 2 = 14 So, (6 + 10) – 2 = 6 + (10 – 2) (b) 16 – (8 – 3) and (16 – 8) – 3 16 – (8 – 3) = 16 – 8 + 3 = 11 (16 – 8) – 3 = 16 – 8 – 3 = 5 So, 16 – (8 – 3) ≠ (16 – 8) – 3 (c) 27 – (18 + 4) and 27 + (– 18 – 4) 27 – (18 + 4) = 27 – 18 – 4 = 5 27 + (– 18 – 4) = 27 – 18 – 4 = 5 So, 27 – (18 + 4) = 27 + (– 18 – 4)

4. In each of the sets of expressions below, identify those that have the same value. Do not evaluate them, but rather use your understanding of terms. (a) 319 + 537, 319 – 537, – 537 + 319, 537 – 319 (b) 87 + 46 – 109, 87 + 46 – 109, 87 + 46 – 109, 87 – 46 + 109, 87 – (46 + 109), (87 – 46) + 109

Answer: (a) 319 – 537 and – 537 + 319 have the same terms So, 319 – 537 and – 537 + 319 have the same value. (b) 87 + 46 – 109, 87 + 46 – 109, 87 + 46 – 109 and also, 87 – 46 + 109 and (87 – 46) + 109 have the same terms and so have equal values.

5. Add brackets at appropriate places in the expressions such that they lead to the

values indicated. (a) 34 – 9 + 12 = 13 (b) 56 – 14 – 8 = 34 (c) – 22 – 12 – 10 + 22 = – 22
Answer: (a) 34 – (9 + 12) = 34 – 21 = 13 (b) (56 – 14) – 8 = 42 – 8 = 34 (c) – 22 – (12 + 10) + 22 = – 22 – 22 + 22 = – 22

More textbook solutions — companion p. 11

6. Using only reasoning of how terms change their values, fill the blanks to make the expressions on either side of the equality (=) equal. (a) 423 + _____ = 419 + _____ (b) 207 – 68 = 210 – _____
Answer: (a) 423 + 419 = 419 + 423 (Commutative Property) (b) 207 – 68 = 207 + 3 – 3 – 68 = 207 + 3 – (3 + 68) = 210 – 71 (Associative Property)
7. Using the numbers 2, 3, and 5, and the operators '+' and ' – ', and brackets, as necessary, generate expressions to give as many different values as possible. For example, 2 – 3 + 5 = 4 and 3 – (5 – 2) = 0
Answer: Some of the expressions are 1. 3 – 2 – 5 = – 4 2. 5 + 2 + 3 = 10 3. 5 + 2 – 3 = 4 4. 2 – (3 + 5) = – 6 Form more expressions.
8. Whenever Jasoda has to subtract 9 from a number, she subtracts 10 and adds 1 to it. For example, 36 – 9 = 26 + 1 (a) Do you think she always gets the correct answer? Why? (b) Can you think of other similar strategies? Give some examples.
Answer: (a) Yes 36 – 9 = 36 – (10 – 1) = 36 – 10 + 1 = 26 + 1 (b) Yes To subtract 8: Subtract 10 and add 2 (since – 8 = – 10 + 2). Ex: 55 – 8 = 55 – (10 – 2) = 55 – 10 + 2 More such strategies can be thought of.

More textbook solutions — companion p. 12

9. Consider the two expressions: (a) 73 – 14 + 1 (b) 73 – 14 – 1 For each of these expressions, identify the expressions from the following collection that are equal to it. (a) 73 – (14 + 1) (b) 73 – (14 – 1) (c) 73 + (– 14 + 1) (d) 73 + (– 14 – 1)
Answer: Expressions (b) and (c) are equal to the expression 73 – 14 + 1, and expressions (a) and (d) are equal to the expression 73 – 14 – 1.

Page No. 39 If another friend, Sangmu, joins them and orders the same items, what will be the expression for the total amount to be paid?
Answer: Expression: 3 × (43 + 24).

Page No. 40 5 × 4 + 3 ≠ 5 × (4 + 3). Can you explain why?
Answer: 5 × 4 + 3 = (5 × 4) + 3 = 23. 5 × (4 + 3) = 5 × 7 = 35 So, 5 × 4 + 3 ≠ 5 × (4 + 3) Is 5 × (4 + 3) = 5 × (3 + 4) = (3 + 4) × 5?
Answer: Yes, all three expressions are equal.

Page No. 41 Use this method to find the following products: (a) 95 × 8 (b) 104 × 15 (c) 49 × 50 Is this quicker than the multiplication procedure you use generally?

More textbook solutions — companion p. 13

Answer: (a) 95 × 8 = (100 – 5) × 8 = 100 × 8 – 5 × 8 = 800 – 40 = 760 (b) 104 × 15 = (100 + 4) × 15 = 100 × 15 + 4 × 15 Try further. (c) 49 × 50 = (50 – 1) × 50 = 50 × 50 – 50 × 1 Try further. Yes, this procedure is quicker than the multiplication procedure we generally use. Which other products might be quicker to find, like the ones above?
Answer: This method is useful when one of the numbers is close to a multiple of 10, 50, 100, 1000, etc. For example: 98 × 7 = (100 – 2) × 7 ; 103 × 9 = (100 + 3) × 9 49 × 5 = (50 – 1) × 5 Find some more!

Page No. 41 – 42
Figure it Out
1. Fill in the blanks with numbers and boxes by signs, so that the expressions on both

sides are equal. () 3 × (6 + 7) = 3 × 6 + 3 × 7 (b) (8 + 3) × 4 = 8 × 4 + 3 × 4 (c) 3 × (5 + 8) = 3 × 5 3 × _____ (d) (9 + 2) × 4 = 9 × 4 2 × _____ (e) 3 × (_____ + 4) = 3 _____ + _____ (f) (_____ + 6) × 4 = 13 × 4 + _____ (g) 3 × (_____ + _____) = 3 × 5 + 3 × 2 (h) (_____ + _____) × _____ = 2 × 4 + 3 × 4

More textbook solutions — companion p. 14

Answer: (c) 3 × (5 + 8) = 3 × 5 3 × 8 (d) (9 + 2) × 4 = 9 × 4 2 × 4 (e) 3 × (10 + 4) = 3×10 + 3 × 4 (f) (13 + 6) × 4 = 13 × 4 + 6 × 4 (g) 3 × (5 + 2) = 3 × 5 + 3 × 2 (h) (2 + 3) × 4 = 2 × 4 + 3 × 4 (i) 5 × (9 – 2) = 5 × 9 – 5 × 2 (j) (5 – 2) × 7 = 5 × 7 – 2 × 7 (k) 5 × (8 – 3) = 5 × 8 5 × 3 (l) (8 – 3) × 7 = 8 × 7 3 × 7 (m) 5 × (12 – 3) = 5 × 12 5 × 3 (n) (15 – 6) × 7 = 15 × 7 6 × 7 (o) 5 × (9 – 4) = 5 × 9 – 5 × 4 (p) (17 – 9) × 7 = 17 × 7 – 9 × 7 [In (e), (m) you may put any number other than 10 and 3 respectively.] + + – – – –

More textbook solutions — companion p. 15

2. In the boxes below, fill in '<', '>' or '=' after analysing the expressions on the LHS and RHS. Use reasoning and understanding of terms and brackets to figure this out, and not by evaluating the expressions. (a) (8 – 3) × 29 (3 – 8) × 29 (b) 15 + 9 × 18 (15 + 9) × 18 (c) 23 × (17 – 9) 23 × 17 + 23 × 9 (d) (34 – 28) × 42 34 × 42 – 28 × 42

Answer: (a) > : since 8 – 3 > 3 – 8 (b) < (c) < (d) = [Try to reason in other parts]

3. Here is one way to make 14: 2 × (1 + 6) = 14. Are there other ways of getting 14? Fill them out below: (a) _____ × (_____ + _____) = 14 (b) _____ × (_____ + _____) = 14 (c) _____ × (_____ + _____) = 14 (d) _____ × (_____ + _____) = 14

Answer: (a) 2 × (5 + 2) = 14 (b) 2 × (3 + 4) = 14 (c) 7 × (1 + 1) = 14 (d) 2 × (6 + 1) = 14

4. Find out the sum of the numbers given in each picture below in at least two different ways. Describe how you solved it through expressions. (I) (II)

More textbook solutions — companion p. 16

Answer: For I: Way1: (5 × 4) + (4 × 8) = 52 Way 2: 2 × (4 + 8 + 4) + (8 + 4 + 8) = 52 For II: Way 1: (8 × 5) + (8 × 6) = 88 Way 2: 8 × (5 + 6) = 88 Find more ways.

Page No. 42 – 44

Figure it Out

1. Read the situations given below. Write appropriate expressions for each of them and find their values. (a) The district market in Begur operates on all seven days of the week. Rahim supplies 9 kg of mangoes each day from his orchard, and Shyam supplies 11 kg of mangoes each day from his orchard to this market. Find the number of mangoes supplied by them in a week to the local district market. (b) Binu earns Rs. 20,000 per month. She spends Rs. 5,000 on rent, Rs. 5,000 on food, and Rs. 2,000 on other expenses every month. What is the amount Binu will save by the end of the year? (c) During the daytime, a snail climbs 3 cm up a post, and during the night, while asleep, accidentally slips down by 2 cm. The post is 10 cm high, and a delicious treat is on top. In how many days will the snail get the treat?

Answer: (a) Expression: 7 × (9 + 11) kg Value: 140 kg (b) Expression: 12 × 20,000 – 12 × (5000 + 5000 + 2000) Amount Binu will save by the end of the year = Rs. 96,000 (c) In the daytime the snail climbs 3 cm up the post and slips down by 2 cm at night. So, the distance climbed by the snail on the post in full day cycle = 3 – 2 = 1 cm ∴ The distance climbed by the snail in 7 days = 7 cm The height of the post is 10 cm. On the 8th day the snail will climb i.e. 7 × (3 – 2) + 3 × 1 = 10 cm

More textbook solutions — companion p. 17

2. Melvin reads a two – page story every day except on Tuesdays and Saturdays. How many stories would he complete reading in 8 weeks? Which of the expressions below describes this scenario? (a) $5 \times 2 \times 8$ (b) $(7 - 2) \times 8$ (c) 8×7 (d) $7 \times 2 \times 8$ (e) $7 \times 5 - 2$ (f) $(7 + 2) \times 8$ (g) $7 \times 8 - 2 \times 8$ (h) $(7 - 5) \times 8$

Answer: Expressions describing the scenario (total stories) = (b) and (g).

3. Find different ways of evaluating the following expressions: (a) $1 - 2 + 3 - 4 + 5 - 6 + 7 - 8 + 9 - 10$ (b) $1 - 1 + 1 - 1 + 1 - 1 + 1 - 1 + 1 - 1$

Answer: (a) Way 1 : $1 - 2 + 3 - 4 + 5 - 6 + 7 - 8 + 9 - 10 = (1 + 3 + 5 + 7 + 9) + (- 2 - 4 - 6 - 8 - 10)$ Way 2: $1 - 2 + 3 - 4 + 5 - 6 + 7 - 8 + 9 - 10 = (1 - 2) + (3 - 4) + (5 - 6) + (7 - 8) + (9 - 10)$ Try more ways. (b) Way 1: $1 - 1 + 1 - 1 + 1 - 1 + 1 - 1 + 1 - 1 = (1 - 1) + (1 - 1) + (1 - 1) + (1 - 1) + (1 - 1)$ Way 2: $1 - 1 + 1 - 1 + 1 - 1 + 1 - 1 + 1 - 1 = (1 + 1 + 1 + 1 + 1) + (- 1 - 1 - 1 - 1 - 1)$ Try more ways.

4. Compare the following pairs of expressions using '<', '>', or '=,' or by reasoning. (a) $49 - 7 + 8$ $49 - 7 + 8$ (b) $83 \times 42 - 18$ $83 \times 40 - 18$ (c) $145 - 17 \times 8$ $145 - 17 \times 6$ (d) $23 \times 48 - 35$ $23 \times (48 - 35)$ (e) $(16 - 11) \times 12 - 11 \times 12 + 16 \times 12$ (f) $(76 - 53) \times 88$ $88 \times (53 - 76)$

More textbook solutions — companion p. 18

Answer: (a) = (b) > (c) < (d) > (e) = (f) > (g) = (h) >

5. Identify which of the following expressions are equal to the given expression without computation. You may rewrite the expressions using terms or removing brackets. There can be more than one expression that is equal to the given expression. (a) $83 - 37 - 12$ (i) $84 - 38 - 12$ (ii) $84 - (37 + 12)$ (iii) $83 - 38 - 13$ (iv) $- 37 + 83 - 12$ (b) $93 + 37 \times 44 + 76$ (i) $37 + 93 \times 44 + 76$ (ii) $93 + 37 \times 76 + 44$ (iii) $(93 + 37) \times (44 + 76)$ (iv) $37 \times 44 + 93 + 76$

Answer: (a) (i) and (iv) (b) (iv)

6. Choose a number and create ten different expressions having that value.

Answer: Let's choose the number 26. Here are ten different expressions with the value 26: 1. $10 + 16$ 2. $30 - 4$ 3. 13×2 4. 52 2 5. $5 + (3 \times 7)$ 6. $(5 \times 5) + 1$ Try for some other numbers.

Illustrated notes: arithmetic expressions

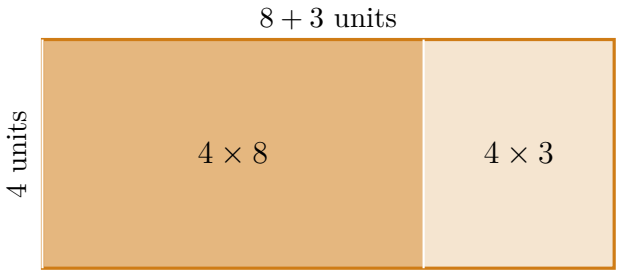
Expressions and terms

An **expression** joins numbers with operations, e.g., $18 - 5 + 4$; its **value** is 17. Rewrite subtraction as addition of a negative: $18 + (-5) + 4$. The signed terms are $18, -5, +4$.

Key Idea

Same value, different expressions: $10 + 2 = 3 \times 4 = 24 \div 2$. These are equal expressions, not identical strings.

Brackets and operations



The area picture shows $4(8 + 3) = 4 \times 8 + 4 \times 3$.

The order of operations

1. Simplify brackets.
2. Multiplication/division, from left to right.
3. Addition/subtraction, from left to right.

Worked Example

$20 - 3 \times 4 + 6 = 20 - 12 + 6 = 14$, whereas $(20 - 3)(4 + 6) = 17 \times 10 = 170$.

The distributive property

$$a(b + c) = ab + ac \qquad a(b - c) = ab - ac.$$

Worked Example

$98 \times 7 = (100 - 2) \times 7 = 700 - 14 = 686$. This is quicker than long multiplication for numbers near 100.

Memory Trick

Think **"the outside number visits every term inside."** $5(12 - 3) = 5 \times 12 - 5 \times 3$.

Three traps to avoid

Common Mistake

1. $10 - (4 + 2) = 4$, but $10 - 4 + 2 = 8$.
2. $7 - 3 + 5$ can be regrouped as $7 + (-3) + 5$, not $7 + (3 + 5)$.
3. $(a + b) \times c$ needs $ac + bc$, not $ac + b$.

Last-minute revision

- Translate stories into expressions before computing.
- Keep signs attached to the terms when rearranging.
- Brackets show what must be taken together.
- Check your result by evaluating the original expression.

Mini-lab: brackets through a story

Child A	Child B	Child C
Notebook + pen	Notebook + pen	Notebook + pen

$$3 \times (20 + 5) = 75$$

Three children buy one Rs. 20 notebook and one Rs. 5 pen each. Calculate $3(20 + 5)$ or $3 \times 20 + 3 \times 5$. Both give Rs. 75. Brackets group one child's total price.

Investigate: always, sometimes, never

- Is $a + b = b + a$ for whole numbers? **Always**.
- Is $a - b = b - a$ for numbers a, b? **Sometimes** (when $a = b$).
- Is $a(b + c) = ab + c$? **Sometimes**, but not generally.
- Is $a - (b + c) = (a - b) - c$? **Always**.

Quick self-check (with answers)

1. $30 - 5 \times 4 = ?$ **10**.
2. $(30 - 5) \times 4 = ?$ **100**.
3. 104×9 mentally $= ?$ **936**, using $(100 + 4)9$.
4. $7(12 - 2) = ?$ **70**, using $84 - 14$.
5. Are $3 + (-5) + 7$ and $3 + 7 - 5$ equal? **Yes**, both 5.
6. Four packs of six pencils and two loose pencils. Write the expression. $4 \times 6 + 2 = 26$.
7. $20 - (7 - 2) = ?$ **15**.
8. Convert $65 - 18 - 12$ to a sum of signed terms. $65 + (-18) + (-12)$.

Memory Trick

When distributing, every term inside the brackets gets multiplied. When you change subtraction into addition, keep the **negative sign**.